06.10.2026

Exercise 8



Goal Today: Time-Domain Specifications, PID

Remember the course objectives

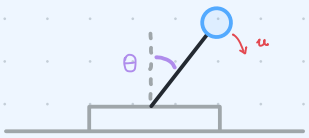
Modeling: represent real-world systems with mathematical equations.

Analysis: understand how a system behaves: input → output

Synthesis: change the system so that it behaves the way we want.

What we did so far:

W1 physical system



W2 modelling

$$ml^2\ddot{\theta} = mgl\sin\theta - c\dot{\theta} + u$$

W3 linearization

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$
$$\begin{pmatrix}\dot{\theta}\\ \ddot{\theta}\end{pmatrix} = \begin{pmatrix}0 & 1\\ 2 & -1\end{pmatrix}\begin{pmatrix}\theta\\ \dot{\theta}\end{pmatrix} + \begin{pmatrix}0\\ 0{,}2\end{pmatrix}u$$
$$y = \begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}\theta\\ \dot{\theta}\end{pmatrix}$$

W4 stability

eigenvalues $-2, +1$

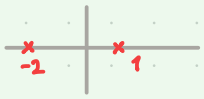
$+1 \Rightarrow$ unstable

W5 transfer function

$$G(s) = \frac{0{,}2}{s^2 + s - 2}$$

W6 poles, zeros

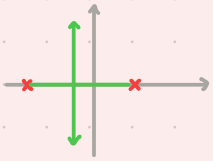
$$G(s) = \frac{0{,}2}{(s+2)\cdot(s-1)}$$



W7 Feedback, root-locus

→ change the poles with feedback

→ see if system is stable



W8 time domain specs

stable is not enough!

how fast? overshoot?

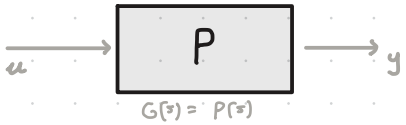
exactly on target?

which K? which C?

Modeling
Analysis
Synthesis

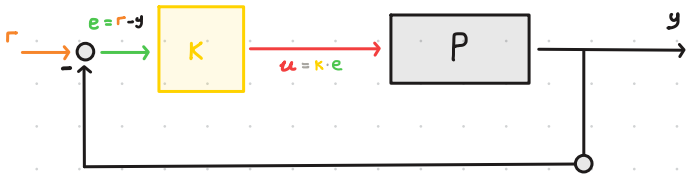
Made with Goodnotes

- Last week we had a system we wanted to stabilize.



$$P(s) = \frac{0,2}{s^2 + s - 2} = \frac{0,2}{\underbrace{(s+2)}_{P_1=-2} \cdot \underbrace{(s-1)}_{P_2=1}}$$

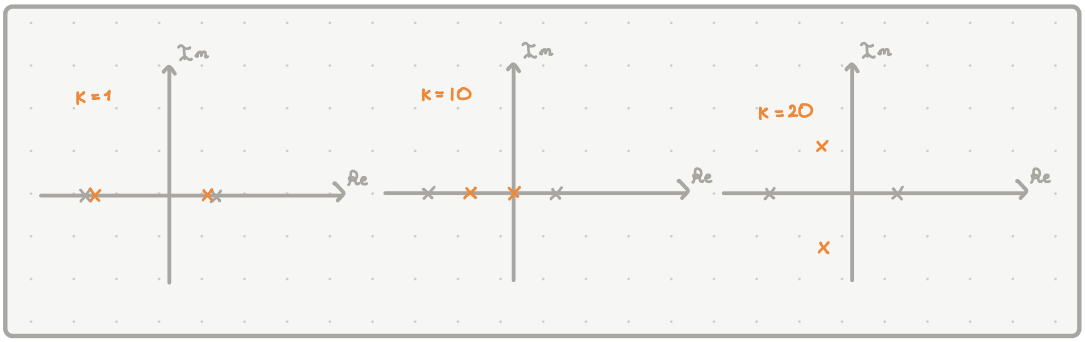
- For this, we used the most basic controller, $C(s) = K$, a proportional controller and we closed the loop with feedback, so that the controller acts on error.



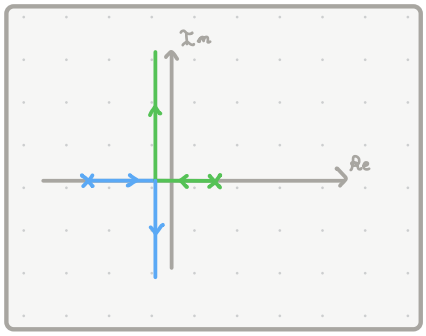
$$T(s) = \frac{K \cdot \frac{0,2}{s^2+s-2}}{1 + K \cdot \frac{0,2}{s^2+s-2}} = \frac{0,2 \cdot K}{s^2 + s - 2 + 0,2K}$$

- Notice how the poles now depend on K. We can adjust K to replace the closed loop poles.

Solve: $s^2 + s - 2 + 0,2K = 0$ for different K:



⇒ We can trace out the paths that the poles take and obtain:

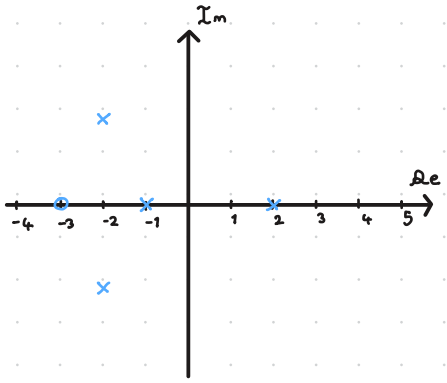


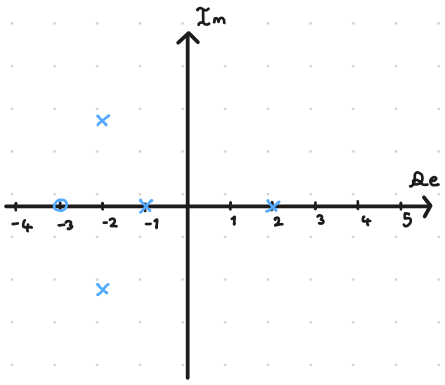
- The plot we just created is called root locus and it shows the location of the closed-loop poles, based on the open loop poles.

- And it allows to quickly see if a controller is feasible or not, like we just did! In this case a simple controller is enough to stabilize the system.

Made with Goodnotes

- Let's look at a few examples again:

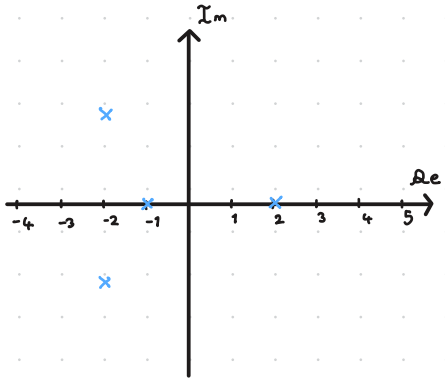


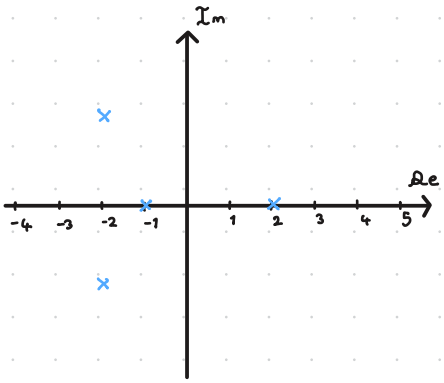


Poles: 4 # Zeros: 1

Asymp. centroid:

Asymp. angle :





Poles: 4 # Zeros: 0

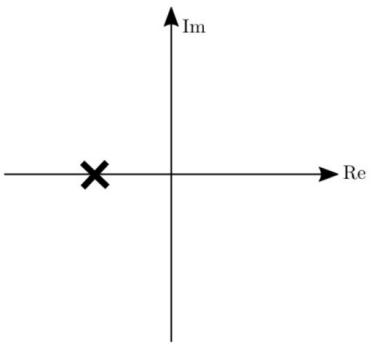
Asymp. centroid:

Asymp. angle :

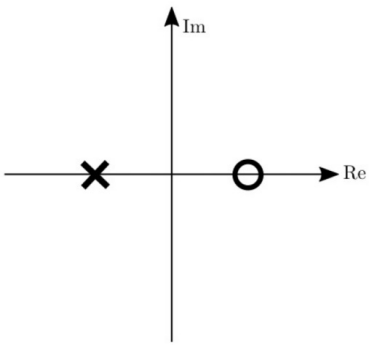
Your Turn:

Question 24 *Mark all correct statements. (2 Points)*

Consider the following pole/zero-diagrams of two plants $P(s)$:



Open-Loop System 1



Open-Loop System 2

Evaluate which of the following statements are correct:

A Any controller $C(s) = k$ with $k > 0$ will stabilize open-loop system *2*.

B A controller $C(s) = \frac{k}{s}$ with $k > 0$ will destabilize open-loop system *2* already for very small gains k.

C Any controller $C(s) = k$ with $k > 0$ will stabilize open-loop system *1*.

D A controller $C(s) = \frac{k}{s}$ with $k > 0$ will destabilize open-loop system *1* already for very small gains k.

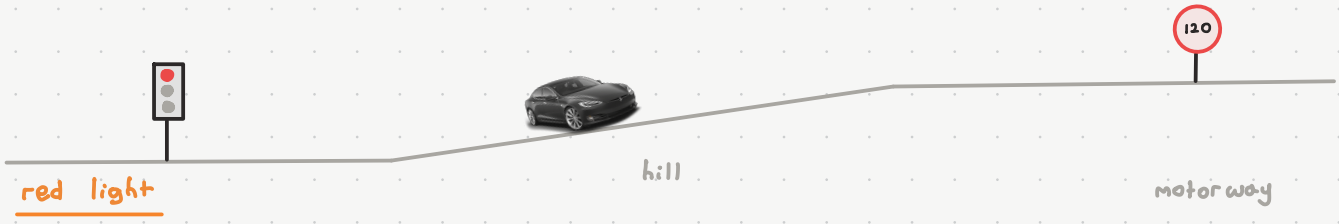
Made with Goodnotes

- So we asked the root locus one question: are all the poles in the left half-plane? Yes or no?

- But a car that doesn't crash is not yet a car you'd buy. Today the system has to be fast, calm and precise. We will read them all From the same picture.

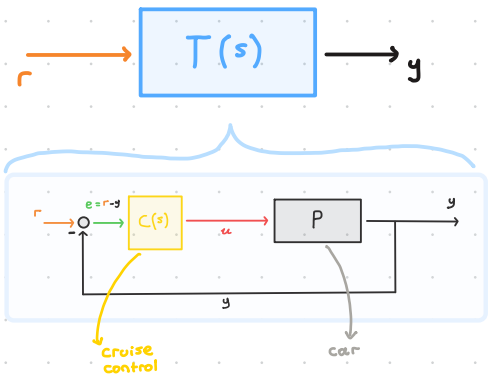
Today's roadmap

We plan on buying a new car with cruise control. We will follow it along its path and see a spec. of the cruise controller at every stage.



0 km The light turns green. The cruise control is set to 50 km/h ($r = 50$). How long until we are actually going 50 km/h?

Our system is closed loop with a closed loop transfer function. Set speed in, actual speed out:



Let's look at a first order system (one pole) as a first approximation of the CL-system

$$T(s) = \frac{1}{\tau \cdot s + 1}$$

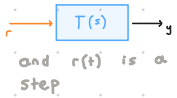
$$\dot{x} = -1/\tau \cdot x + 1/\tau \cdot u$$
$$y = x$$

pole at $-1/\tau$

- We can calculate $y(t)$ with our formula from week 4.

$$y(t) = C \cdot e^{A \cdot t} \cdot \cancel{x_0}^{0} + C \cdot \int_0^t e^{A(t-\tau)} \cdot B \cdot \underbrace{u(\tau)}_{r(\tau) = 50} d\tau + \cancel{D}^{0} u(t)$$

$$y(t) = 50 \cdot (1 - e^{-t/\tau})$$



and r(t) is a step

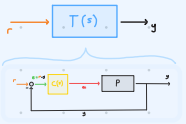
Made with Goodnotes

Our question:

How much time to get to 50 km/h?

Our model:

$$T(s) = \frac{1}{\tau s + 1}$$

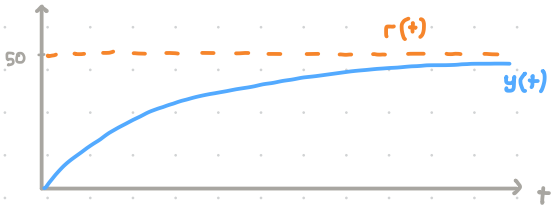


Our time response:

$$y(t) = 50 \cdot (1 - e^{-t/\tau})$$

- So let's answer the question!

$$y(t) = 50$$
$$1 - e^{-t/\tau} = 1$$
$$e^{-t/\tau} = 0 \Rightarrow \text{uh oh}$$



⇒ The first order system never reaches 50!

⇒ Because of this, we reformulate our question and define settling time T_d.

Our question:

How much time to ~~get to~~ 50 km/h?

⇒

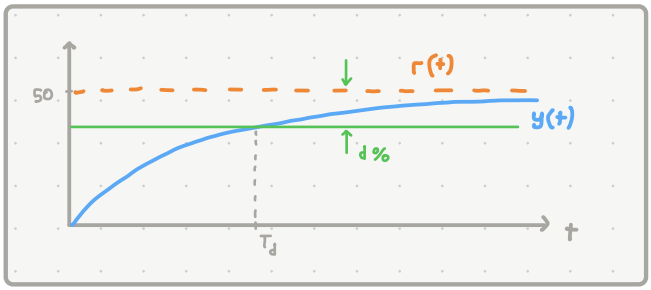
Our question:

How much time to get within d% of 50 km/h (the steady state)

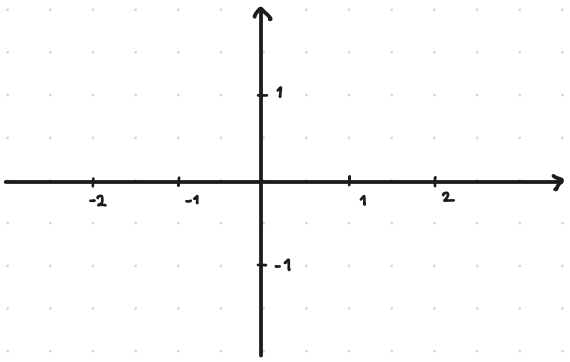
⇒

$$T_d = \tau \ln(100/d)$$

$$y(t) = r \cdot (1 - d/100)$$
$$1 - e^{-t/\tau} = 1 - d/100$$
$$t = \tau \cdot \ln(100/d)$$

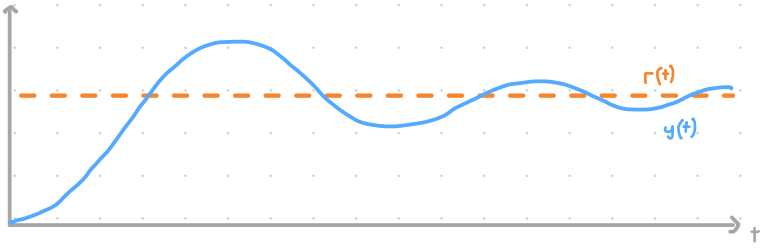


- Now suppose we want $T_d \leq 3$ for $d = 5$. We can use this to set requirements on our τ and thus our poles.



Made with Goodnotes

- But in real life, usually something else happens:

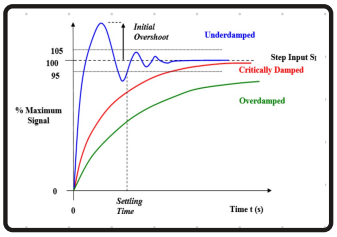


- The cruise control gets there, goes past, resettles, and so on...
- Such behavior is represented with a second-order system (2 poles)

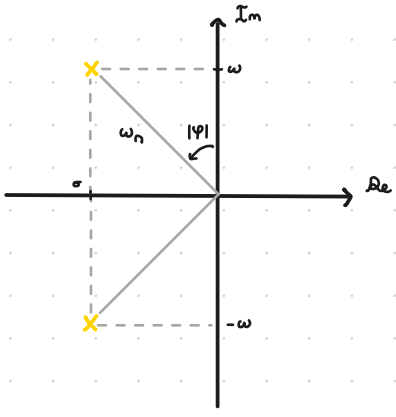
$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} x + \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} u$$

$$y = x$$



- We can map out the poles located at $s = \sigma + j\omega$ in the complex plane and their relation to the system parameters:



$$\omega_n = \sqrt{\sigma^2 + \omega^2}$$

$$|\sigma| = \zeta\omega_n$$

$$\varphi = \arctan\left(\frac{\sigma}{\omega}\right)$$

$$\zeta = \sin(|\varphi|)$$

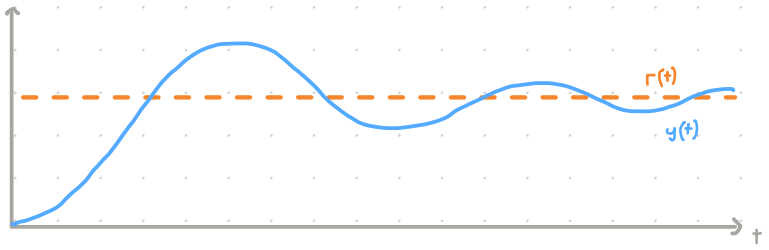
- For an underdamped system ($\zeta < 1$) with zero initial condition we can find the time response to a step like we did earlier for the first-order system

$$y(t) = C \cdot e^{A \cdot t} \cdot \cancel{x_0}^{0} + C \cdot \int^{t} e^{A(t-\tau)} \cdot B \cdot \underbrace{u(\tau)}_{r(\tau) = 50} d\tau + \cancel{D u(t)}^{0}$$

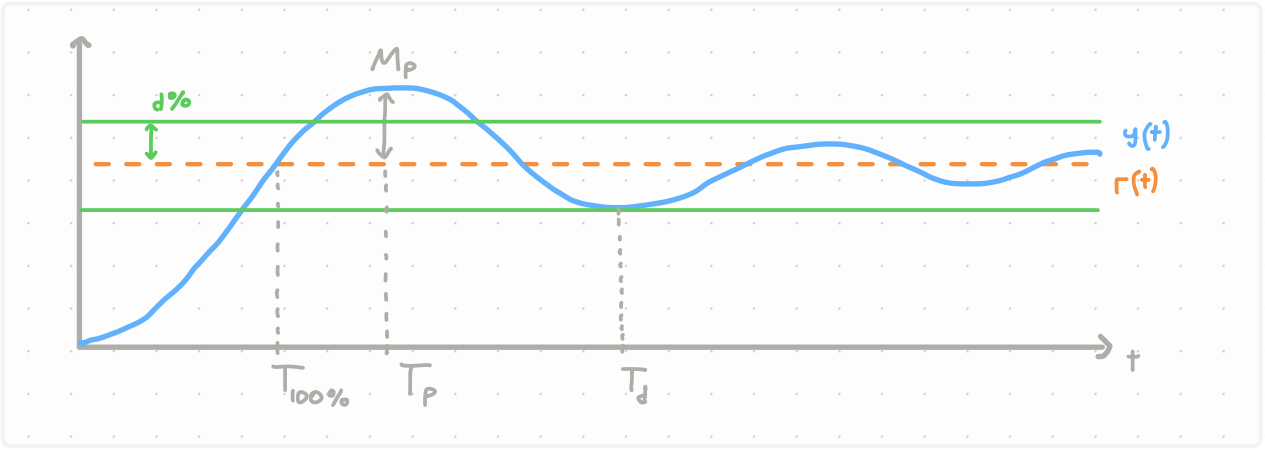
$$y(t) = 1 - \frac{1}{\cos\varphi} \cdot e^{\sigma \cdot t} \cdot \cos(\omega \cdot t)$$

Made with Goodnotes

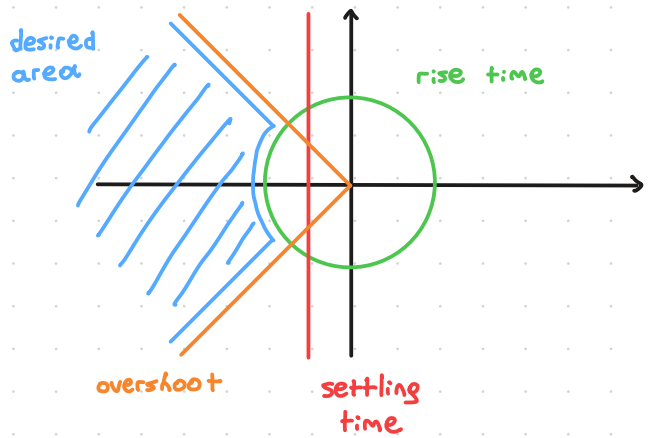
- Because of the shape of the response, we can ask it way more questions than we did to the 1st ord. system:



Our question:	Our question:	Our question:	Our question:
How much time to reach 50?	How much time to peak?	What is the peak overshoot?	How much time to get within d% of 50?
⇓	⇓	⇓	⇓
Rise time	Time to peak	Peak overshoot	Settling time
$T_{100\%} = \frac{\pi/2 - \varphi}{\omega} \approx \frac{\pi}{2\omega_n}$	$T_p = \frac{\pi}{\omega}$	$M_p = e^{\frac{\sigma \cdot \pi}{\omega}}$	$T_d = \frac{1}{\lvert\sigma\rvert} \cdot \ln\left(100/d\right)$



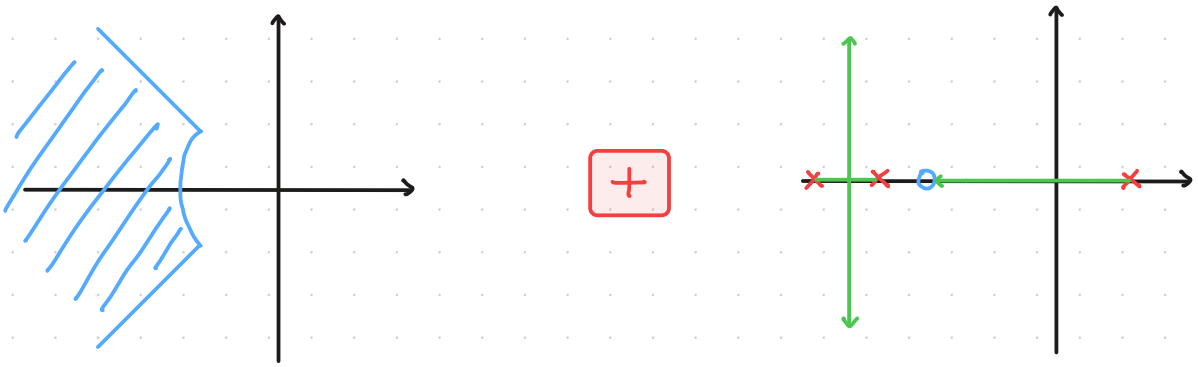
- This now means we can again, take requirements in the time domain and transform them into constraints on the location of the poles of the system

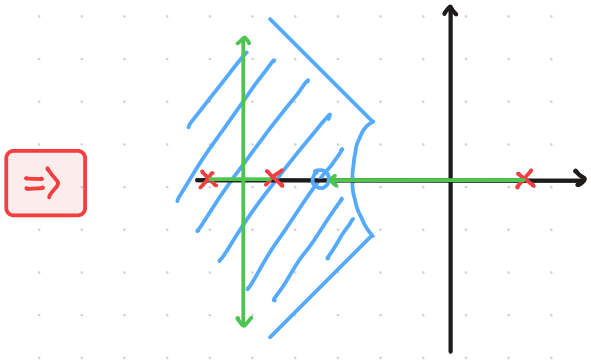


⇒ This gives us a clear requirement on where we should try to get our poles to!

Made with Goodnotes

- So now that we have a clear region in the s-domain we want to aim for, the natural next step is to see if a proportional controller can get us there using a root locus plot:



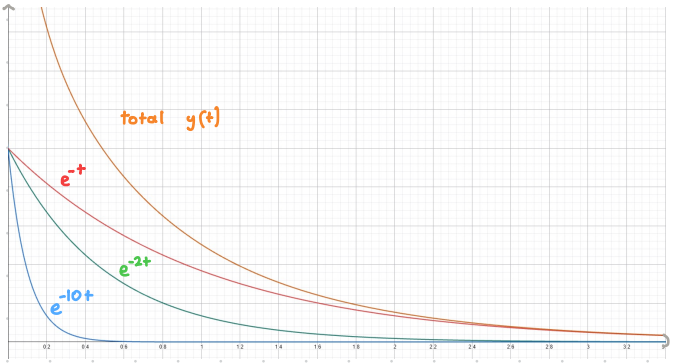


⇒ On this example, we see that a p-controller is enough to meet our demands at some K.

Dominant Pole Approximation

- We modeled the car as a first and second order system, but what if actually it was a higher order system? (>2 poles)
- In that case we use the dominant pole approximation.

⇒ Suppose our system is: $G(s) = \frac{1}{(s+2)\cdot(s+10)\cdot(s+1)}$ which produces following terms in the time response:



- We see that $e^{-10\cdot t}$, the fastest pole goes to zero relatively fast and becomes irrelevant.
- We can thus approximate this system (or any system >2 poles) with the slowest poles. (poles closest to the Im axis)

Made with Goodnotes



- We see that they match almost exactly!
- We can approx. this system as

$$G(s) = \boxed{\frac{1}{10}}.$$

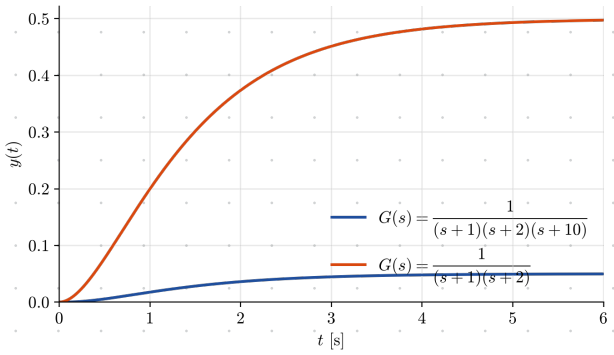
- **Watch out!**: When doing dominant pole approximation, the steady-state value must remain the same.

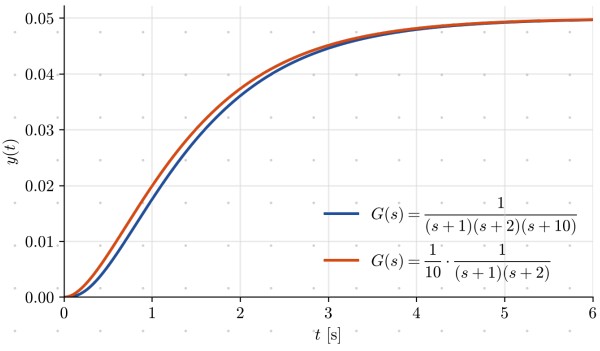
⇒ Remember ss-response: $e^{st} \rightarrow \boxed{G(s)} \rightarrow G(s) \cdot e^{st}$

Our input is a constant step: $r = r \cdot e^{0 \cdot t}$

$r \rightarrow \boxed{G(0)} \rightarrow G(0) \cdot r$

⇒ This means $G(0) \stackrel{!}{=} G(0)_{dom}$. That's why we added the $\boxed{\frac{1}{10}}$.





Your turn

Box 7: Questions 34, 35

You are considering the following transfer function:

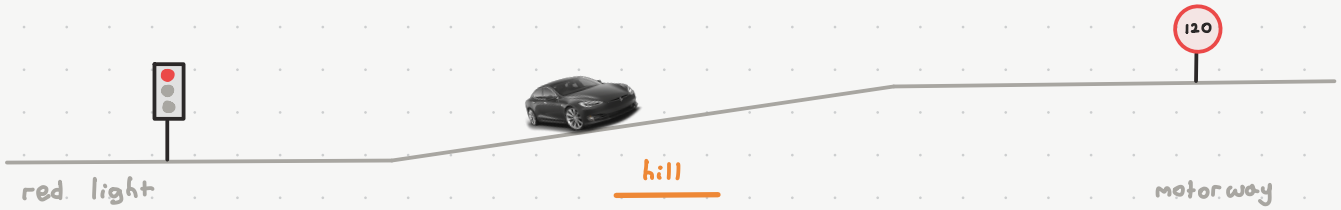
$$L(s) = \frac{(s+1)\left(s+\frac{1}{100}\right)}{\left(s+\frac{1}{2}\right)(s+3)(s+4)(s+30)}$$

Question 34 *Choose the correct answer. (1 Point)*

To simplify the calculations, you approximate the transfer function with a simpler model. Select the approximation which can reasonably be assumed to model the behavior of $L(s)$ for low frequencies ω with negligible error.

- ☐ $L(s) \approx \frac{1}{3000} \cdot \frac{(s+1)}{\left(s+\frac{1}{2}\right)(s+3)(s+4)}$
- ☐ $L(s) \approx \frac{1}{30} \cdot \frac{(s+1)}{\left(s+\frac{1}{2}\right)(s+3)(s+4)}$
- ☐ $L(s) \approx \frac{1}{30} \cdot \frac{(s+1)\left(s+\frac{1}{100}\right)}{\left(s+\frac{1}{2}\right)(s+3)(s+4)}$
- ☐ $L(s) \approx \frac{(s+1)\left(s+\frac{1}{100}\right)}{\left(s+\frac{1}{2}\right)(s+3)(s+4)}$

Made with Goodnotes

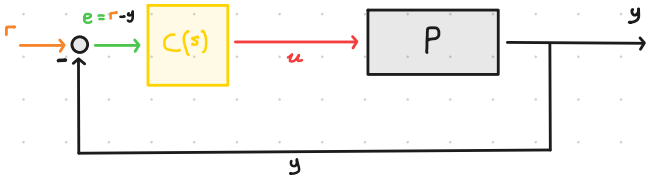


hill On the flat the car ended exactly at 50 km/h. Now the road tilts up: the car slows down. The car pushes, but the speedometer settles at 48 km/h. Why is it short? How can we fix it?

- From a few weeks ago we know the final value theorem:

$$y_{ss} = \lim_{t\to\infty} y(t) = \lim_{s\to 0} s\cdot Y(s) = \lim_{s\to 0} s\cdot G(s)\cdot U(s)$$

to determine the ss-output of a system. Can we do something similar with the CL-system?



- Let's try to compute the ss-error. We already defined the TF from $r \to e$ last week as $S(s) = \frac{1}{1+L(s)}$, i.e. $E(s) = S(s)\cdot R(s)$. We can now write:

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to 0} s\cdot S(s)\cdot R(s) = \lim_{s\to 0} s\cdot \frac{1}{1+L(s)}\cdot R(s)$$

Looking at $L(s)$ in Bode-Form we see:

$$L(s) = \frac{K_{Bode}}{s^{\delta}}\cdot\frac{\left(\frac{s}{-z_1}+1\right)\cdot\ldots\left(\frac{s}{-z_m}+1\right)}{\left(\frac{s}{-p_1}+1\right)\cdot\ldots\left(\frac{s}{-p_{n-s}}+1\right)} \xrightarrow{s\to 0} L(0) = \frac{K_{Bode}}{s^{\delta}}$$

δ: no. of poles at origin (integrators)

if we now take a step reference and $L(0)\to\infty$

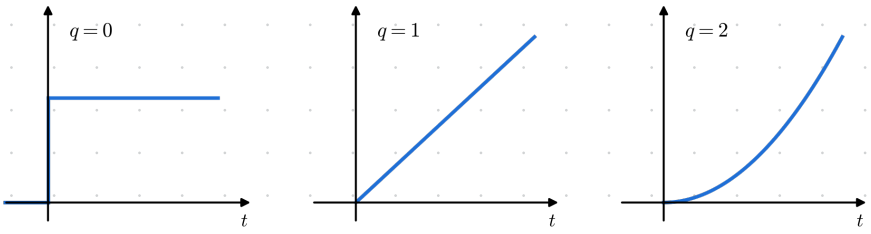
then $\lim_{s\to 0} s\cdot\frac{1}{1+L(s)}\cdot\frac{1}{s} = e_{ss} \longrightarrow 0$

Made with Goodnotes

- That means in order to obtain $e_{ss} = 0$ to a step reference we need a pole at the origin, an integrator.
- IF we choose another reference we potentially need more than 1.

$$r(t) = \frac{1}{q!} t^q \quad \text{or in the } s\text{-domain} \quad R(s) = \frac{1}{s^{q+1}}$$

q: order of the ramp



- Depending on the order of the ramp and type of system (no. of integrators) we get different ss-errors.

e_{ss}	$q = 0$	$q = 1$	$q = 2$
Type 0	$\frac{1}{1 + k_{\text{Bode}}}$	∞	∞
Type 1	0	$\frac{1}{k_{\text{Bode}}}$	∞
Type 2	0	0	$\frac{1}{k_{\text{Bode}}}$

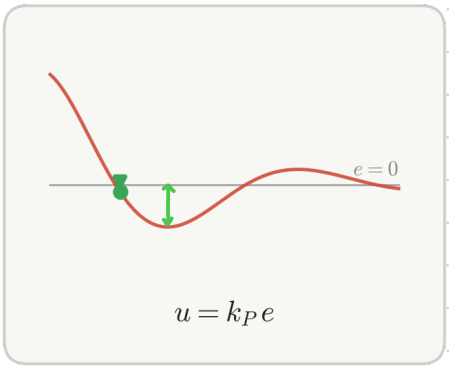
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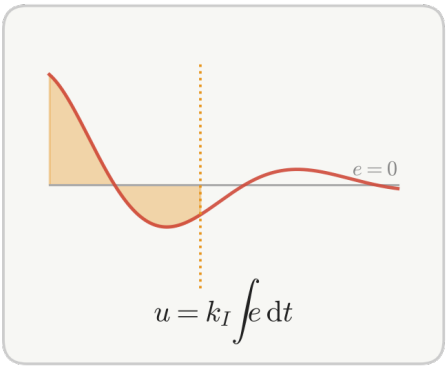
motorway The limit goes to 120, there are hills, and the passanger wants a smooth ride. Now we build the real controller.

The controller is a driver. It sees one thing, the error $e(t)$ and it commands an input accordingly. We can read this error in three ways:

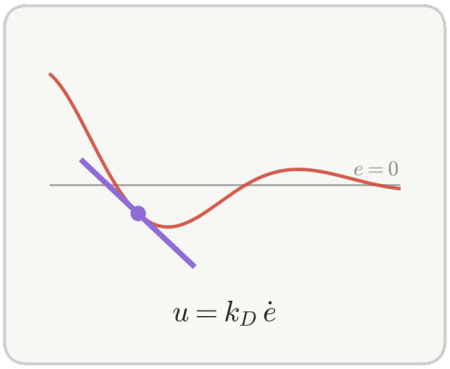
P looks at the gap now



I remembers every gap so far



D watches the gap closing

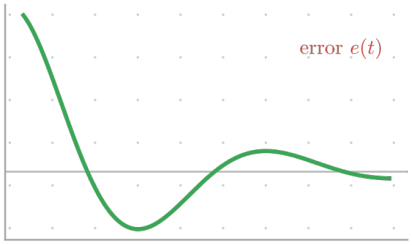


=> P: press proportional to the err.

=> I: keep a running total of the err.

=> D: watch how fast the err is growing.

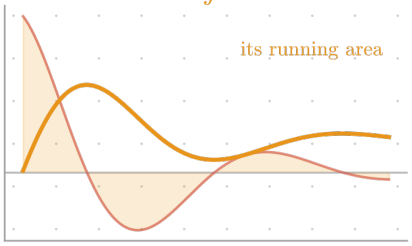
P: $k_P e(t)$



⇓ s-domain

K_P

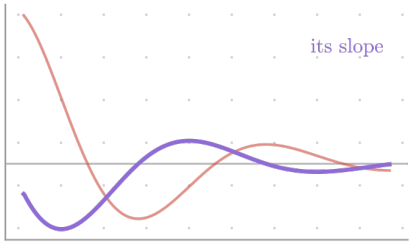
I: $k_I \int e \, dt$



⇓ s-domain

$K_I \cdot 1/s$

D: $k_D \dot{e}(t)$



⇓ s-domain

$K_D \cdot s$

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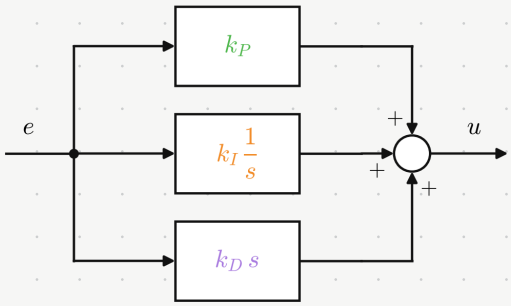
This is called PID control:

P → Proportional

I → Integral

D → Derivative

} each term handles the error differently



$$C(s) = K_P + K_I \cdot 1/s + K_D \cdot s$$

The effects of each can be summarized:

Proportional k_P

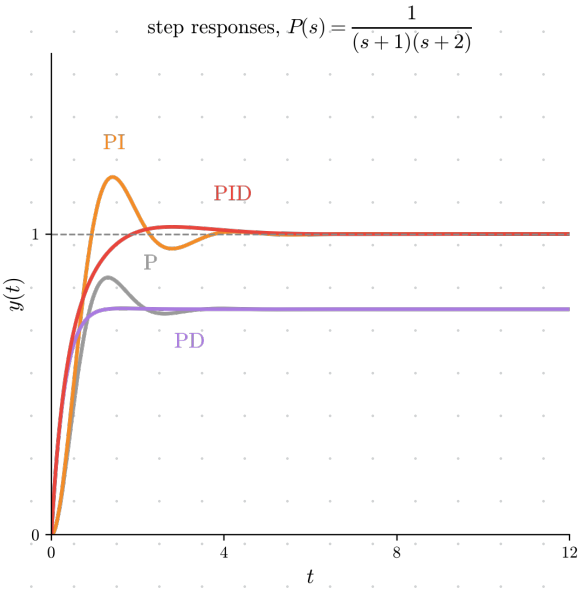
- decreases the steady-state error (never to zero)
- increases the closed-loop bandwidth: faster
- increases the sensitivity to noise
- can reduce the stability margin of high-order systems

Integral $k_I\,/\,s$

- eliminates the steady-state error to a step (if the closed loop is stable)
- reduces the stability margins

Derivative $k_D\, s$

- reduces overshoot, increases damping
- improves the stability margins
- increases the sensitivity to noise



Your Turn

Question 29 *Mark all correct statements. (2 Points)*

Mark all the correct statements about the behavior of PID controllers.

- ☐ The derivative action can be highly sensitive to measurement noise, thus it is often fed with a filtered error signal.
- ☐ Inserting an integral action always helps to achieve better tracking error.
- ☐ As the derivative gain increases, the steady-state error is not affected.
- ☐ As the proportional gain increases, the phase margin increases.
- ☐ The derivative action amplifies overshoots in the system.
- ☐ High proportional gain does not affect the sensitivity to noise

Made with Goodnotes